



TEMPLATE-DESIGNED ORGANIC ELECTRONICS
RESEARCH TRAINING GROUP 2591

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Date of submission: February 4, 2026

C2: Ordering and reaction of quinacridone and derivatives on metal surfaces

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1 State of the art & preliminary work

Surfaces play a crucial role in various scientific and technological fields, including catalysis, sensor technology, and the development of organic electronics. The interaction between organic molecules and metal surfaces is of particular interest due to its implications for the design and optimization of organic electronics.^[1] This exactly matches to one of the main interests of the TiDE project.^[2]

One organic molecule that has been extensively studied in the context of organic electronics is quinacridone (QA), which structure is shown in Figure 1.1. QA is an organic, prochiral molecule with a π -conjugated system, that is known for its use as a dye in printer toners and varnishes.^[3] Its potential extends beyond its application as a dye, as previous studies demonstrated the application of QA in organic electronics.^[4–6] Specific applications for QA have been identified, including its use in organic light-emitting diodes (OLEDs),^[7–9] organic photovoltaics (OPVs)^[10,11] and organic field-effect transistors (OFETs).^[12–15] In addition, QA has been found to form well-defined structures on metal surfaces, such as the Ag(100), the Ag(111), and the Cu(111) surface.^[16–20]

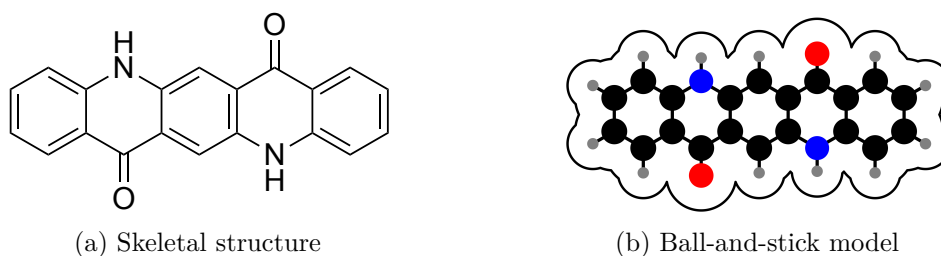


Figure 1.1: Molecular structure of quinacridone (QA).

A common structural motif of QA monolayers on surfaces^[17,20–22] and in its bulk structure^[23] are 1D molecular chains, which form due to intermolecular hydrogen bonds.^[16,17]

In particular, QA on the Ag(100) surface forms two different phases, the α -phase and the β -phase. The α -phase consists of the previously described homochiral, parallel lying 1D molecular chains. This leads to a point-on-line (POL) structure for the α -phase. In contrast, the β -phase is characterized by a commensurate phase, which is thermodynamically more stable than the α -phase and forms irreversibly when the sample is heated. The β -phase has fewer hydrogen bonds and stronger adsorbate-substrate interactions.

The reaction from the α - to the β -phase and their differences are of particular interest, as the reaction involves a partly deprotonation of the amine groups of the QA molecules on the Ag(100) surface and a drastic change in the binding situation of the different atoms,

including a change of the handedness of the molecule. This is not yet fully understood and will be further investigated in this project.

1.1 Preliminary work: XPS and NIXSW measurements of quinacridone on the Ag(100) surface

As said above, QA on the Ag(100) surface forms two structural phases: the α - and the β -phase. The arrangements of both phases are expected to be driven by intermolecular hydrogen bonds between the keto group of one molecule with the amine group of another molecule and adsorbate-substrate interactions.^[16–18]

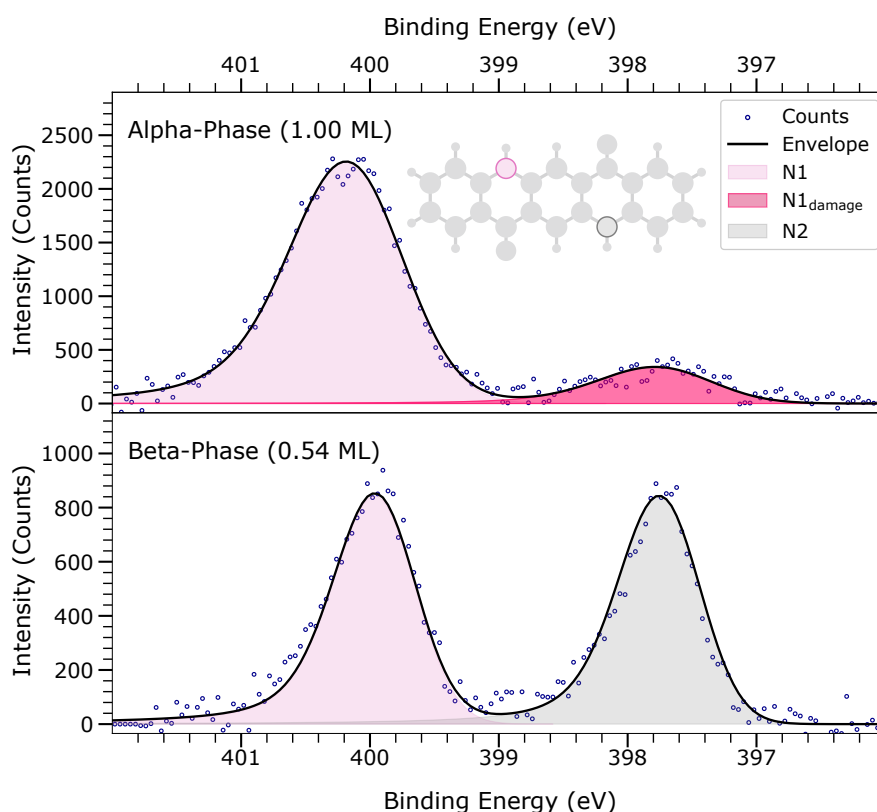


Figure 1.2: N1s XPS spectra of quinacridone (QA) on the Ag(100) surface for the α - and β -phases, showing the different components, the envelope and the data points.^[24]

As the knowledge about these two phases was originally limited to scanning tunneling microscopy (STM) and low energy electron diffraction (LEED) measurements, which only give information about the structural arrangement, x-ray photoelectron spectroscopy (XPS) and normal incidence x-ray standing wave (NIXSW) measurements were lately

performed to gain insight into the electronic structure and adsorption height of QA on the Ag(100) surface.^[24,25]

Figure 1.2 shows the XPS measurements that revealed significant differences in the C1s, O1s, and especially N1s core-level spectra between the two phases. This indicates drastic changes in the chemical environment of the N- and O-atoms upon the phase transition. The amine groups of the QA molecules undergo a partly deprotonation during the phase transition from the α - to the β -phase, as evidenced by the emergence of a second peak in the N1s XPS at lower binding energies.^[24]

Furthermore, the NIXSW measurements provided valuable information about the adsorption heights of the N-atoms in both phases, revealing a smaller adsorption height of the deprotonated N-atoms on the Ag(100) surface in the β -phase. This shows a stronger interaction between the deprotonated N-atoms and the Ag(100) surface in the β -phase compared to the α -phase. In contrast, the C- and O-atoms exhibit only a minor decrease in their adsorption heights from the α to the β -phase. The N1s NIXSW spectra for both phases are illustrated in Figure 1.3.^[25]

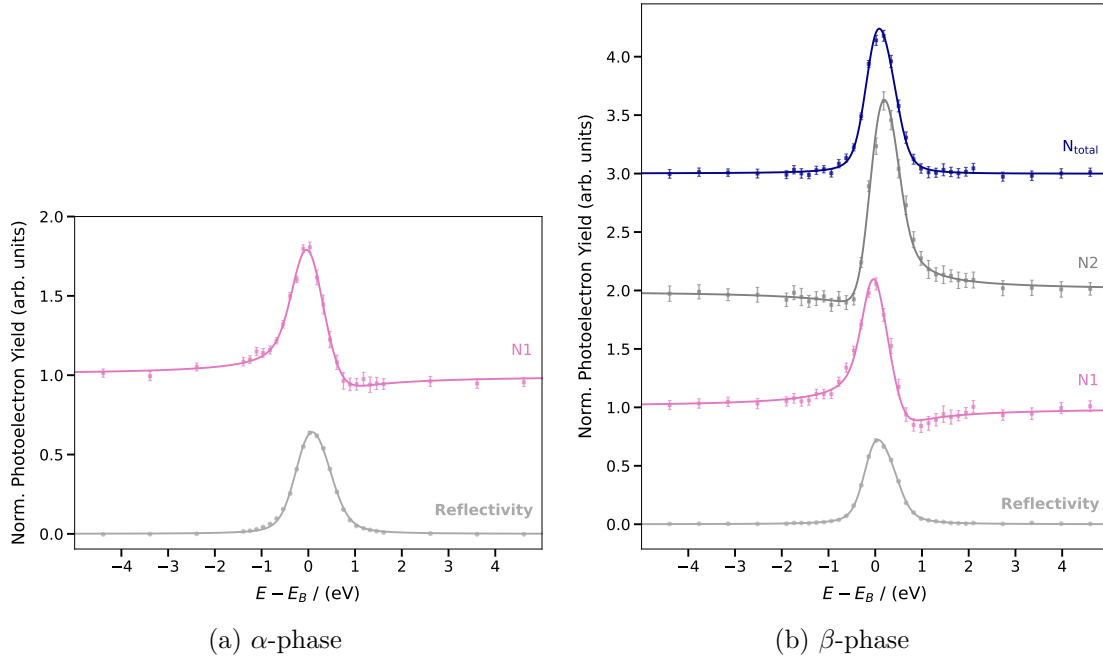


Figure 1.3: N1s NIXSW spectra of quinacridone (QA) on the Ag(100) surface for the (a) α - and (b) β -phases, showing the different components, the envelope and the data points.^[25]

The XPS and NIXSW measurements result in a revised structure model of the β -phase of QA on the Ag(100) surface. The structures are based on the LEED and STM results

from literature^[16,17] and the new insights from the XPS and NIXSW measurements. The revised structure models for both phases are illustrated in Figure 1.4.

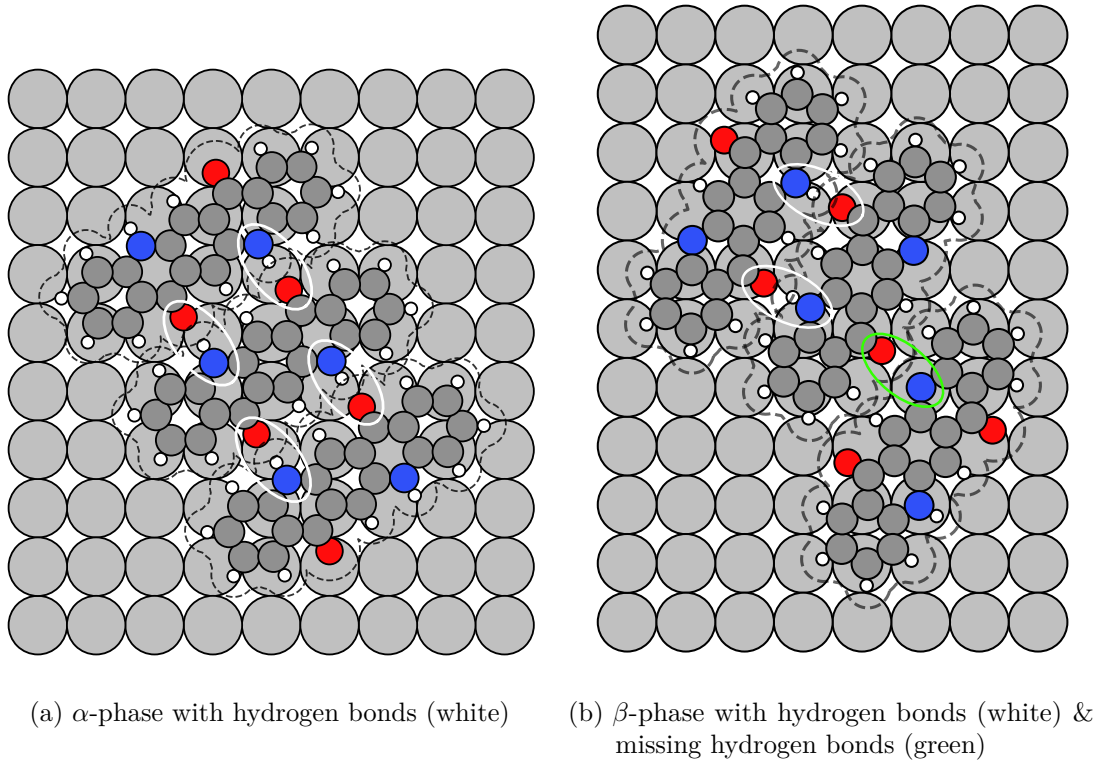


Figure 1.4: Structure model of the α - and β -phases of QA on the Ag(100) surface.^[25]

The major change in this structure model is the deprotonation of one N-atom per QA molecule in the β -phase, which leads to a stronger interaction with the Ag(100) surface and a reduced adsorption height. This revised structure model provides a more accurate representation of the adsorption behavior of QA on the Ag(100) surface in both phases and highlights the significant changes that occur during the phase transition. However, further investigation are necessary to fully understand the mechanisms driving the phase transition and the associated structural changes, such as the partly deprotonation or the change in the handedness of 50 % of the molecules.

2 Research vision & project idea

The research vision of this project is to gain a comprehensive understanding of the adsorption behavior, surface reactions, and structural properties of QA and its derivatives on metallic surfaces. The key objectives of this project are twofold: one is to investigate the effect of molecular modifications on the ordering of QA derivatives on metallic surfaces, and the other is to further explore the reaction of QA on the Ag(100) surface, particularly focusing on the deprotonation and the change in the handedness of the molecules during the phase transition from the α - to the β -phase.

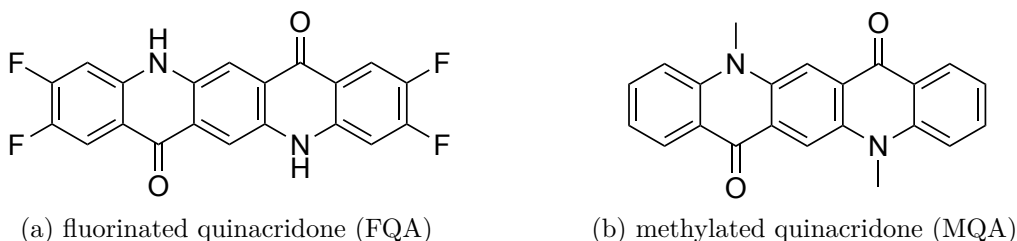


Figure 2.1: Molecular structures of FQA and MQA.

Part 1: Influence of molecular modifications on adsorption behavior and film growth

This includes investigating the influence of molecular modifications on the adsorption characteristics and film growth. FQA and MQA are of particular interest due to their altered electronic properties and potential impact on intermolecular interactions and surface interactions.

For example, the presence of F-atoms in FQA introduces strong electronegativity, which can affect the adsorption behavior and ordering on metallic surfaces. The position of the F-atoms at the ends of the molecule will introduce electronegative substituents between the 1D-linear chains of QA, having a significant impact of the intermolecular chain distances. In addition, the F-atoms can influence the molecule-substrate interactions due to their high electronegativity, which may influence the phase behavior.

This can be directly compared to the known behavior of QA on the same surface, allowing an interpretation of the effect by the F-atoms. Similar investigations on fluorinated organic molecules on metallic surfaces have shown significant changes, e.g. for Peryleno[3,4-cd:9,11-c'd']dipyran-3,5,10,12-tetrone (PTCDA).^[26]

On the other side, MQA introduces steric hindrance due to the presence of methyl groups on the N-atoms, which will affect the ordering of the molecules via hydrogen

bonds. There are already QA derivatives with long alkyl groups on the N-atoms, showing hydrogen bonding in their crystal structures.^[27,28] Investigating MQA on metallic surfaces will provide insights into how altered hydrogen bonding influence the ordering of MQA compared to QA.

Part 2: Reaction of quinacridone with metallic surfaces

The second part of the project focuses on the phase transition of QA on the Ag(100) surface. The XPS and NIXSW measurements have already revealed significant changes in the chemical environment and adsorption heights of the N-atoms upon the phase transition from the α - to the β -phase. However, the detailed mechanisms underlying this phase transition and the associated structural changes are not yet fully understood.

The phase transition involves two major changes, that need to be further investigated: the partly deprotonation of the amine groups of the QA molecules and the change in the handedness of 50 % of the molecules.

This investigation will involve equipping the ultra-high vacuum (UHV) chamber containing the STM with a heating stage to enable controlled temperature variations during the measurements. This allows the observation of the structural changes that occur during the phase transition to a preset temperature. Furthermore, one may be able to observe intermediate states during the phase transition, providing information about the rearrangement process.

Another interesting aspect of this part is to investigate whether similar phase transitions occur for FQA or even MQA on the Ag(100) surface or other metallic surfaces, such as Cu(111) and Au(111). This will provide insights into the influence of molecular modifications and surface properties on the phase transition behavior of QA and its derivatives.

Especially the phases for MQA are of interest, as the methyl groups attached to the N-atoms will definitely effect the deprotonation of the reaction from the α - to the β -phase observed for QA.

Applied techniques

The project will primarily utilize STM and LEED techniques to investigate the adsorption behavior, film growth, and phase transition of QA and its derivatives on metallic surfaces. STM will provide details about the molecular arrangement and ordering on the surfaces, while LEED will provide complementary information about the surface unit cell and periodicity.

In addition, XPS and NIXSW techniques may be employed to gain detailed information into the electronic structure and chemical environment of the molecules on the surfaces. A beamtime at the Diamond Light Source (DLS) must be used for these techniques. This

will either focus on the FQA properties on the Ag(100) surface to allow a comparison to the QA molecule or the triangulation of the β -phase of QA on the Ag(100) surface to further investigate the deprotonation process.

Context in the TiDE project

This project perfectly matches with the Focus Area C: Surfaces and Interfaces. The proposed research will contribute to this focus area by providing insights into the adsorption behavior and film growth of QA and its derivatives on metallic surfaces, as well as the influence of molecular modifications on these processes.

Furthermore, there are synergies with other projects within the TiDE initiative, such as the Focus Area A: Molecular Building Blocks, which e.g. synthesizes molecules, that may be used as an adsorbate. In addition, the Focus Area E: Theory & Modeling, would be able to support the interpretation and validation of the experimental results through theoretical calculations.

3 Project milestone & working plan for the next 12 months

The idea is to systematically investigate the adsorption behavior and film growth of QA and its derivatives on various metallic surfaces, including Ag(100), Cu(111), and Au(111). The project will be divided into several key milestones, illustrated in Figure 3.1.

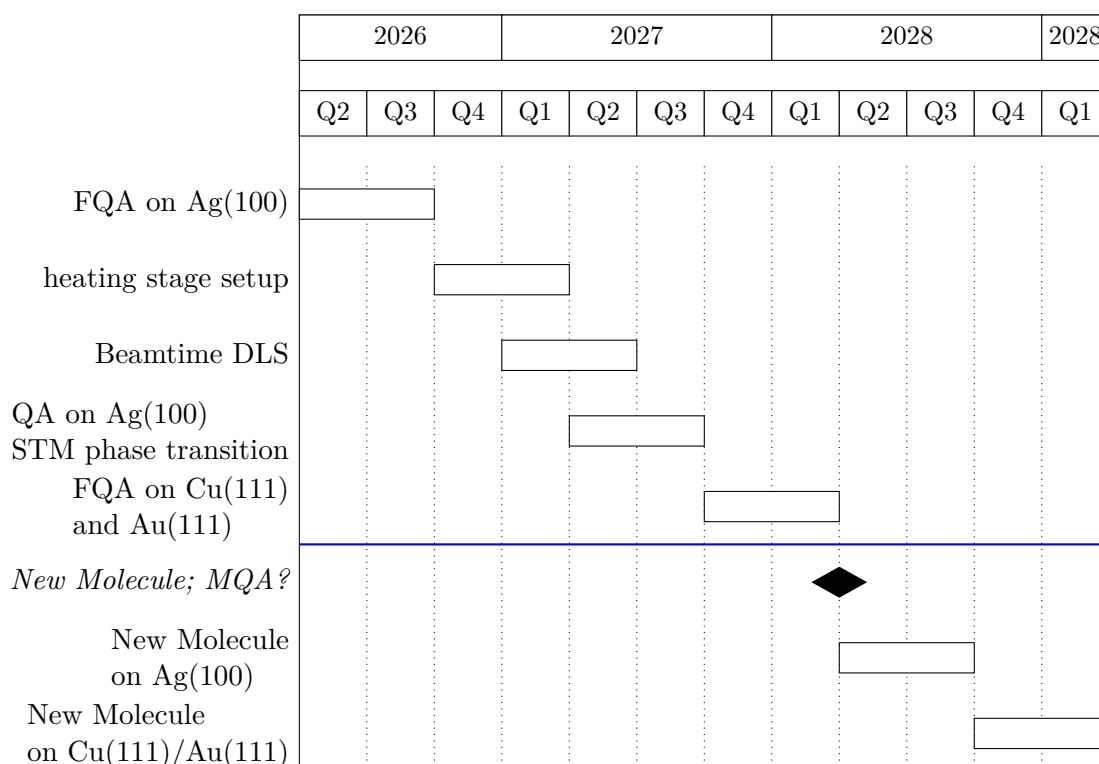


Figure 3.1: Gantt chart illustrating the project working plan.

The first milestone is the investigation of the phase behavior of FQA on the Ag(100) surface using STM and LEED techniques, providing insights into the structural arrangement of FQA molecules on the Ag(100) surface. These results are comparable with the known behavior of QA on the Ag(100) surface, allowing an interpretation of the effect by the F-atoms.

The second milestone is the detailed study of the phase transition of QA on the Ag(100) surface. This will involve temperature-dependent STM measurements to elucidate the structural changes occurring during the phase transition from the α - to the β -phase.

Therefore, the UHV chamber containing the STM will be equipped with a heating stage to enable controlled temperature variations during the measurements, which are necessary for the phase transition.

These two milestones are planned to be completed within the first 12 months of the project. Afterwards, the project will continue with further investigations into the adsorption behavior of QA derivatives on the Ag(100) and other metallic surfaces, as well as exploring the effects of different deposition conditions on film growth and ordering. Which molecule is investigated next will depend on the results of the first two milestones and will be decided in consultation with the supervisors. A possible candidate is MQA, which is quinacridone with methyl groups attached to the N-atoms.

4 Appendix

List of abbreviations

DLS Diamond Light Source

FQA fluorinated quinacridone

LEED low energy electron diffraction

MQA methylated quinacridone

NIXSW normal incidence x-ray standing wave

OFET organic field-effect transistor

OLED organic light-emitting diode

OPV organic photovoltaic

POL point-on-line

PTCDA Peryleno[3,4-cd:9,11-c'd']dipyran-3,5,10,12-tetrone

QA quinacridone

STM scanning tunneling microscopy

UHV ultra-high vacuum

XPS x-ray photoelectron spectroscopy

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